

KARTHEEK MUKKAVILLI

Austin, TX | 832-752-3951 | kartheek.mukkavilli@gmail.com | linkedin.com/in/kartheek-mukkavilli | github.com/mkskart | U.S. Citizen

EDUCATION

University of Texas at Austin

Austin, TX

B.S. Electrical & Computer Engineering | GPA: 3.95/4.0 | Expected May 2028

Coursework: Embedded Systems · Digital Logic Design · Digital Signal Processing · Discrete Mathematics · Software Design · Circuit Theory

Elements of Computing Certificate (CS) | Business Foundations Minor

TECHNICAL SKILLS

Languages: Python · C++ · Java · SQL · TypeScript

Quant/ML: PyTorch · scikit-learn · Isolation Forest · Genetic Algorithms · Transformer Models · Time-Series Analysis · Backtesting

Infra/Tools: Alpaca API · Groq API · FastAPI · Docker · SQLAlchemy · AWS IoT Core · Git

Math: Linear Algebra · Discrete Mathematics · Digital Signal Processing · Probability & Statistics

EXPERIENCE

Human Centered Robotics Laboratory — UT Austin

Austin, TX

Research Assistant

Jan 2026 – Present

- Trained a PyTorch Transformer on high-frequency glove telemetry to predict 16-DOF bionic hand motion, achieving 1ms sensor-to-actuation latency — directly analogous to low-latency signal-to-execution pipelines.
- Applied PID control and real-time signal processing across 16 joints, improving tracking consistency from unreliable to stable — demonstrates quantitative systems tuning under real-time constraints.

National Oilwell Varco

Houston, TX

Embedded Systems Intern

Jun – Aug 2025

- Architected a 40-node distributed IoT sensor network with reliable data ingestion pipelines to AWS IoT Core — experience applicable to market data collection and distributed systems design.
- Replaced \$20K/site legacy system with \$250 custom board (98% cost reduction); automated deployment tooling cutting onboarding from hours to under one minute across 50 field units.

J.P. Morgan Chase

Houston, TX

Software Engineering Intern

Jun – Jul 2022

- Built a Spring Boot/Cassandra microservice for a 2M+ record production keyspace; optimized SQL query logic reducing latency by 40%.
- Delivered a ReactJS + AG Grid real-time metadata dashboard for 200+ internal engineers on a 4-person scrum team.

PROJECTS

QuantClaw — Algorithmic Trading Engine | Python · Alpaca · Isolation Forest · Groq API

- Genetic algorithm breeds 50 VWAP+RSI strategy variants over 50 generations daily, selecting the fittest bot for live 5-min bar trading under a T+1 cash-settlement risk manager.
- 12-month backtest: 70% win rate over 250 trades, +20% net return. Live paper trading: +\$160 on \$800 account in one month.
- Isolation Forest monitors real-time volatility; autonomous Groq LLM engine makes discretionary trade decisions under hard-coded risk rules.

kinematic-transformer | Python · PyTorch

- Transformer encoder (~69K params) forecasting multi-joint kinematic sequences; val MSE 0.003108, 0.571ms inference latency vs LSTM 0.656ms — latency advantage for real-time control.

LiteWing — Real-Time Control | C++ · FreeRTOS · ESP32

- 1 kHz PID loop with measured 45µs jitter — demonstrates real-time systems discipline applicable to low-latency execution infrastructure.

J.A.R.V.I.S. — Agentic AI System | Python · FastAPI · Ollama · ChromaDB

- Local-first voice AI with RAG memory pipeline (Mem0 + ChromaDB), tool-use across Gmail/Calendar/browser/Telegram, and a FastAPI + WebSocket dashboard.

LEADERSHIP

UT IEEE RAS Engineer | TSA Co-Founder: 3× National Finals | 4× Gold PVSA Medalist | SAMPADA Violin Certified · ABRSM Grade 5